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$$\begin{aligned}\int \sin^2 x \cos^2 x dx &= \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right) dx \\&= \int \frac{1 + \cos 2x - \cos 2x - \cos^2 2x}{4} dx \\&= \int \frac{1 - \cos^2 2x}{4} dx = \int \frac{\sin^2 2x}{4} dx = \frac{1}{4} \int \sin^2 2x \\&= \frac{1}{4} \int \frac{1 - \cos 4x}{2} dx = \frac{1}{8} \int dx - \frac{1}{8} \int \cos 4x dx \\t = 4x &\Rightarrow dx = \frac{dt}{4} \\&\Rightarrow \frac{1}{8} x - \frac{1}{32} \sin 4x + C\end{aligned}$$

References